

[Figure 1](#) provides the prototypical case of the problem of induction as encountered in real life. You observe a hypothetical variable for one thousand days. It could be anything (with a few mild transformations): book sales, blood pressure, crimes, your personal income, a given stock, the interest on a loan, or Sunday attendance at a specific Greek Orthodox church. You subsequently derive *solely from past data* a few conclusions concerning the properties of the pattern with projections for the next thousand, even five thousand, days. On the one thousand and first day—boom! A big change takes place that is completely unprepared for by the past.

Consider the surprise of the Great War. After the Napoleonic conflicts, the world had experienced a period of peace that would lead any observer to believe in the disappearance of severely destructive conflicts. Yet, surprise! It turned out to be the deadliest conflict, up until then, in the history of mankind.

Note that after the event you start predicting the possibility of other outliers happening locally, that is, in the process you were just surprised by, *but not elsewhere*. After the stock market crash of 1929 half of America's traders braced for another one every October—not taking into account that there was no antecedent for the first one. We worry too late—ex post. Mistaking a naïve observation of the past as something definitive or representative of the future is the one and only cause of our inability to understand the Black Swan.

It would appear to a quoting dilettante—i.e., one of those writers and scholars who fill up their texts with phrases from some dead authority—that, as phrased by Hobbes, “from like antecedents flow like consequents.” Those who believe in the unconditional benefits of past experience should consider this pearl of wisdom allegedly voiced by a famous ship's captain:

*But in all my experience, I have never been in any accident... of any sort worth speaking about. I have seen but one vessel in distress in all my years at sea. I never saw a wreck and never have been wrecked nor was I ever in any predicament that threatened to end in disaster of any sort.*

E. J. Smith, 1907, Captain, RMS *Titanic*

Captain Smith's ship sank in 1912 in what became the most talked-about shipwreck in history.<sup>\*</sup>

### ***Trained to Be Dull***

Similarly, think of a bank chairman whose institution makes steady profits over a long time, only to lose everything in a single reversal of fortune. Traditionally, bankers of the lending variety have been pear-shaped, clean-shaven, and dress in possibly the most comforting and boring manner, in dark suits, white shirts, and red ties. Indeed, for their lending business, banks hire dull people and train them to be even more dull. But this is for show. If they